

RQ Answers

omitted

2024-04-05

Data overview

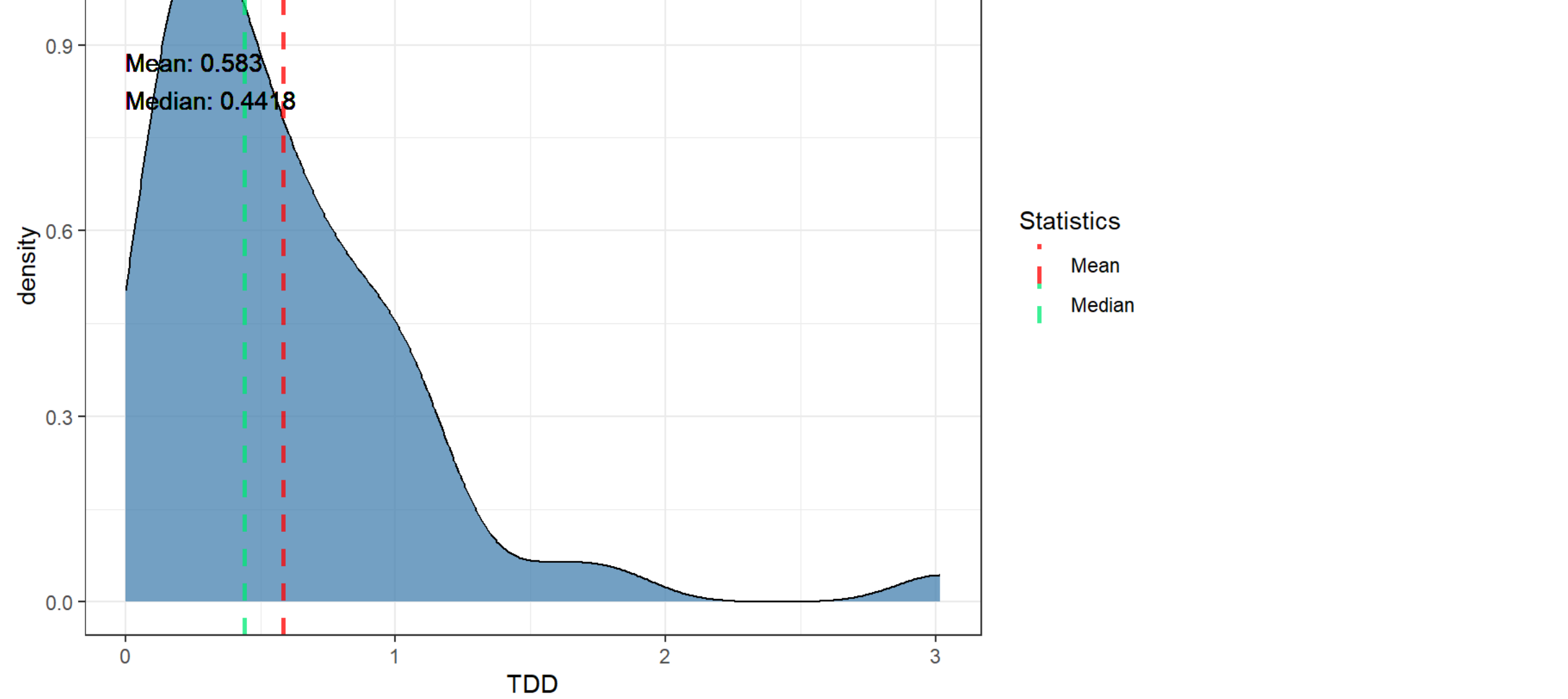
glimpse(all_data)				
## Rows: 6,164				
## Columns: 11				
## \$ repo	<chr>	"commons-lang", "commons-lang", "commons-lang", "c...		
## \$ pr_number	<int>	1033, 1033, 1033, 1033, 1033, 1033, 1033, 10...		
## \$ rule	<chr>	"java:S1192", "java:S3400", "java:S1192", "java:S1...		
## \$ severity	<chr>	"CRITICAL", "MINOR", "CRITICAL", "MAJOR", "MINOR", ...		
## \$ file	<chr>	"src/test/java/org/apache/commons/lang3/reflect/Me...		
## \$ type	<chr>	"CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM...		
## \$ status	<chr>	"OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O...		
## \$ debt	<int>	22, 5, 16, 5, 5, 5, 5, 8, 5, 5, 8, 5, 10, ...		
## \$ ncloc_affected_file	<int>	907, 907, 907, 907, 907, 907, 907, 907, 907, ...		
## \$ complexity	<int>	150, 150, 150, 150, 150, 150, 150, 150, 150, ...		
## \$ origin	<chr>	"PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P...		

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the data has 55 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but the data has 55 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

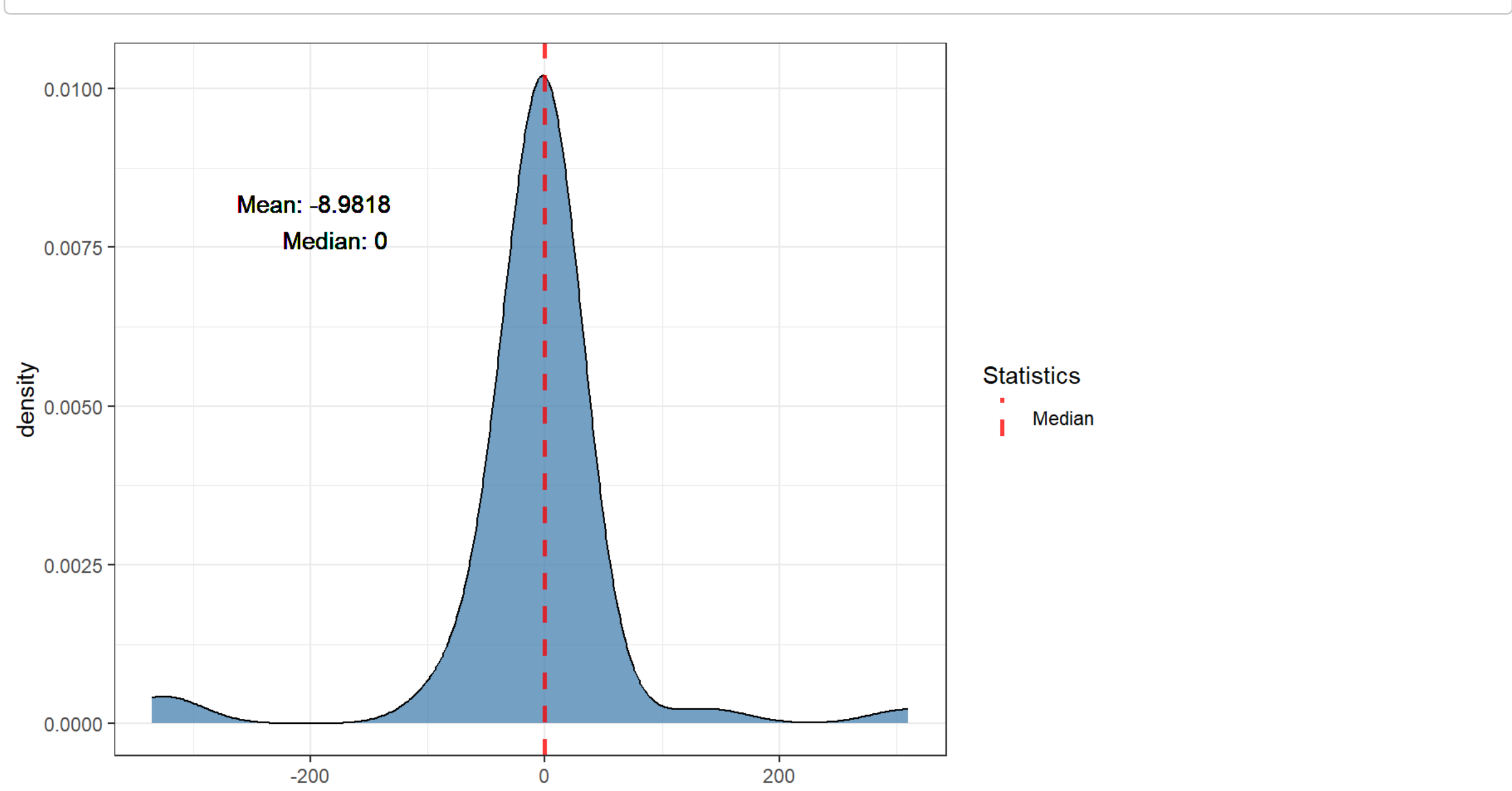


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the data has 55 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but the data has 55 rows.
## # Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
<chr>	<dbl>	<dbl>	<dbl>	
1 commons-lang	21.8	63.6	14.6	

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3				
mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero		
<dbl>	<dbl>	<dbl>	<dbl>	
1	21.8	63.6	14.6	

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3				
repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero		
<chr>	<dbl>	<dbl>	<dbl>	
1 commons-lang	69.1	30.9		

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2				
preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero			
<dbl>	<dbl>	<dbl>	<dbl>	
1	69.1	30.9		

Percentages for new TDV by repo

## # A tibble: 1 × 3				
Groups: repo [1]				
repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero		
<chr>	<dbl>	<dbl>	<dbl>	
1 commons-lang	84.2	15.8		

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2				
mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero			
<dbl>	<dbl>	<dbl>	<dbl>	
1	84.2	15.8		

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 × 2				
## # Groups: rank [10]				
## rank `commons-lang`				
<int>	<chr>			
1	1 java:S4165	-	56	
2	2 java:S1192	-	55	
3	3 java:S125	-	44	
4	4 java:S2293	-	35	
5	5 java:S3878	-	27	
6	6 java:S1612	-	25	
7	7 java:S100	-	16	
8	8 java:S1611	-	9	
9	9 java:S4968	-	6	
10	10 java:S2129	-	6	

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

## # A tibble: 10 × 2				
rule	position_score			
<chr>	<dbl>			
1 java:S4165	122			
2 java:S1192	123			
3 java:S125	124			
4 java:S2293	125			
5 java:S3878	126			
6 java:S1612	127			
7 java:S100	128			
8 java:S1611	129			
9 java:S2129	130			
10 java:S3984	131			

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2				
## # Groups: rank [10]				
## rank `commons-lang`				
<int>	<chr>			
1	1 java:S1192	-	1611	
2	2 java:S100	-	782	
3	3 java:S4165	-	369	
4	4 java:S1172	-	306	
5	5 java:S125	-	306	
6	6 java:S3878	-	233	
7	7 java:S108	-	230	
8	8 java:S3400	-	190	
9	9 java:S1104	-	186	
10	10 java:S1186	-	114	

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

## # A tibble: 10 × 2				
rule	position_score			
<chr>	<dbl>			
1 java:S1192	122			
2 java:S100	123			
3 java:S4165	124			
4 java:S1172	125			
5 java:S125	126			
6 java:S3878	127			
7 java:S108	128			
8 java:S3400	129			
9 java:S1104	130			
10 java:S1186	131			

Outliers

Higher TDD

## # A tibble: 55 × 5				
repo	pr_number	total_debt	total_ncloc	tdd
<chr>	<int>	<int>	<int>	<dbl>
1 commons-lang	1051	978	324	3.02
2 commons-lang	661	1471	813	1.81
3 commons-lang	1151	1356	878	1.54
4 commons-lang	680	1048	895	1.17
5 commons-lang	1033	1046	907	1.15
6 commons-lang	949	1297	1178	1.10
7 commons-lang	407	886	831	1.07
8 commons-lang	237	2640	2613	1.01
9 commons-lang	369	2892	2995	0.966
10 commons-lang	403	256	268	0.955
## # i 45 more rows				

Higher TDV

## # A tibble: 55 × 3				
pr_number	repo	tdv		
<int>	<chr>	<int>		
1	661 commons-lang	310		
2	680 commons-lang	141		
3	228 commons-lang	55		
4	1047 commons-lang	16		
5	949 commons-lang	6		
6	367 commons-lang	5		
7	688 commons-lang	5		
8	1151 commons-lang	5		
9	214 commons-lang	0		
10	216 commons-lang	0		
## # i 45 more rows				

Lower TDV

## # A tibble: 55 × 3				
pr_number	repo	tdv		
<int>	<chr>	<int>		
1	1148 commons-lang	-335		
2	354 commons-lang	-316		
3	899 commons-lang	-86		
4	403 commons-lang	-81		
5	372 commons-lang	-60		
6	369 commons-lang	-40		
7	348 commons-lang	-35		
8	368 commons-lang	-25		
9	1051 commons-lang	-22		
10	234 commons-lang	-20		
## # i 45 more rows				

Extra

Number of PRs with pre-existing TD

## n				
1	55			

Number of java:S1192 issues that affect test code

## n				
1	1666			

Number of java:S1192 issues that affect not test code

## n				
1	0			

Summary NCLOC

## ncloc				
Min.	:	20.0		
1st Qu.	:	394.5		
Median	:	878.0		
Mean	:	1698.0		
3rd Qu.	:	2130.0		
Max.	:	12160.0		